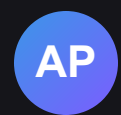


• Microsoft Build AI Hackathon — Agent Swarm

Lokaal

Privacy-first proximity networking, planned by an agent swarm.

A **4-agent AI swarm** on Azure AI Foundry that plans safe, specific, real-world meetups.

 Built solo by **Akash Pradhan**, Software Engineer
github.com/akashpradhan439

One prompt is fast. It is also unaccountable.

BEFORE Single LLM call

"Plan a meetup" → one model, one shot

✗ Generic output

"Grab a coffee sometime" — no person, no place, no time.

✗ Hallucinated venues

Recommends places that do not exist or aren't between anyone.

✗ No safety gate

Nothing reviews the suggestion before it reaches a user.

✗ No audit trail

You can't see why the model said what it said.

AFTER 4-agent swarm

Plan → Research → Execute → Critique, with override

✓ Grounded in real data

Cites only real connections and real nearby venues.

✓ Adversarial Critic

A red-team agent rejects vague or ungrounded suggestions.

✓ Content-Safety checked

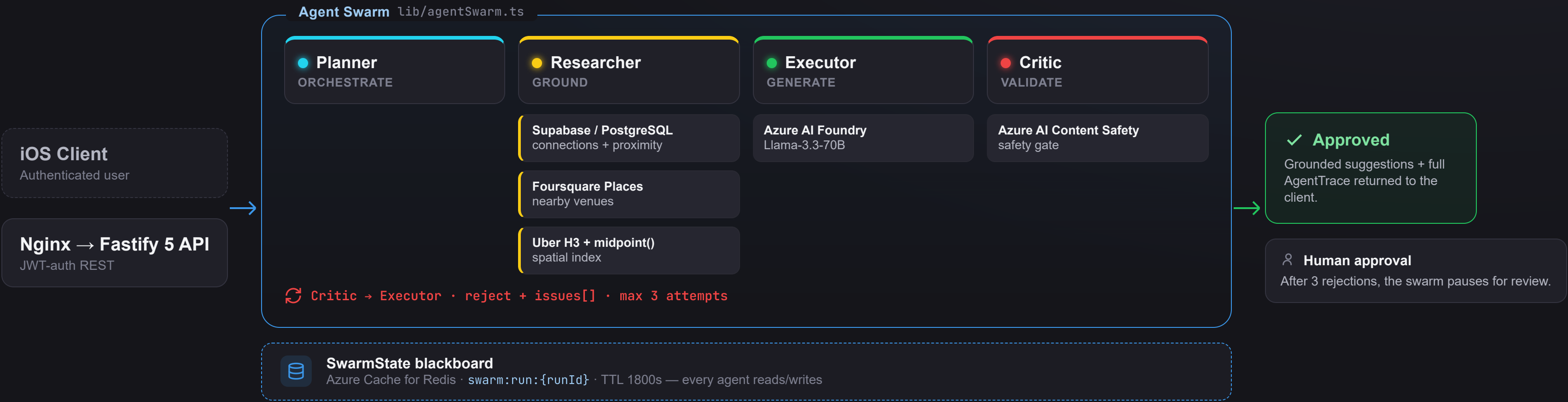
Azure AI Content Safety integrated at the Critic layer.

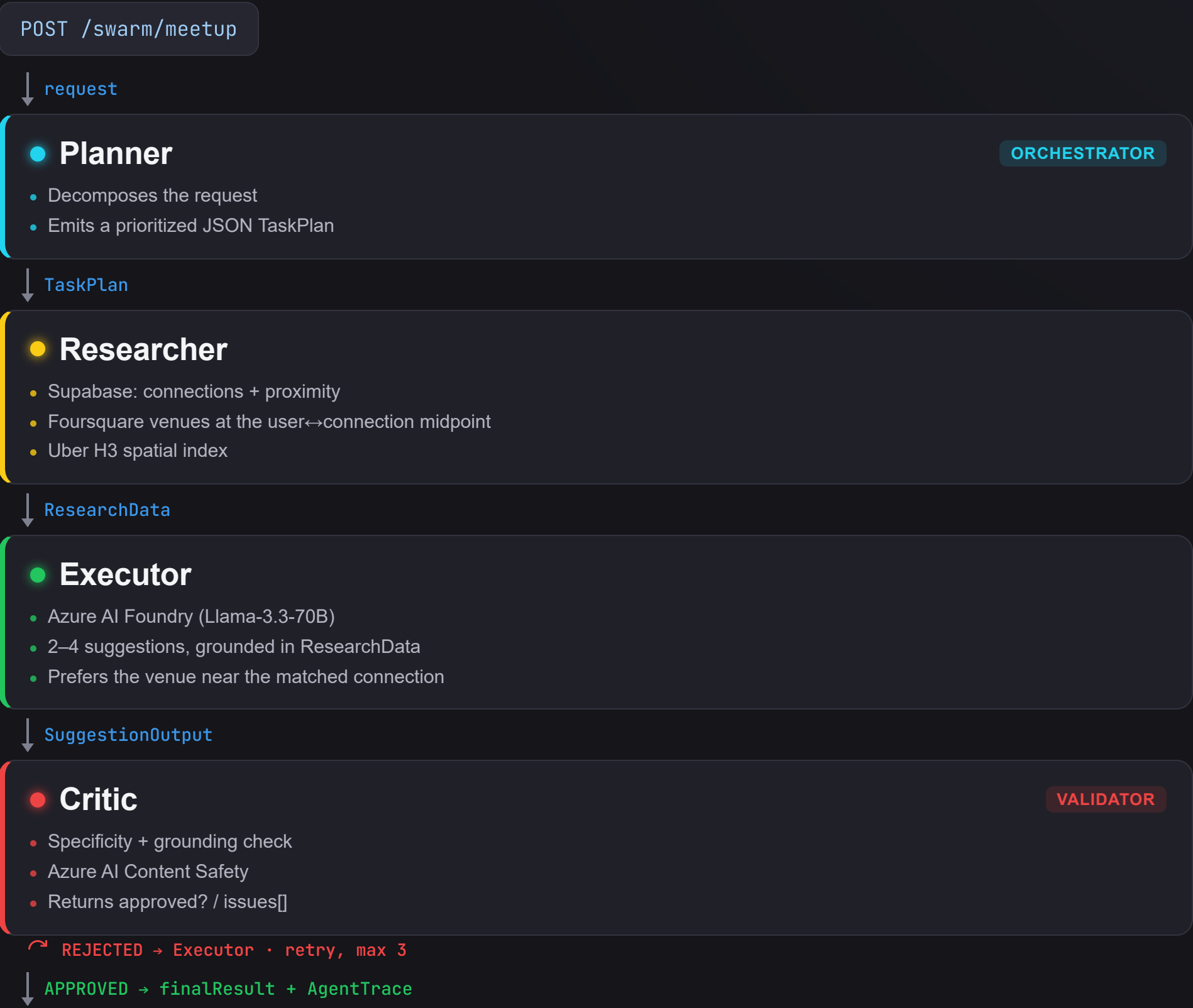
✓ Full trace + human override

Per-step audit log; the swarm pauses for approval when unsure.

Lokaal turns a risky single prompt into an auditable four-agent pipeline — **Plan**, **Research**, **Execute**, **Critique** — that produces meetup suggestions grounded in real data and cleared by an adversarial safety check.







Shared blackboard

```
// Redis · swarm:run:{id} · TTL 1800s
SwarmState {
  runId, phase, plan,
  research, executorOut,
  trace[] // per-step AgentTrace
}
```

One grounding primitive

The same `midpoint()` helper powers **both** the swarm's Researcher and the conversational assistant — venue searches always center on the user↔connection midpoint. Two AI surfaces, one shared primitive.

Human-in-the-loop

After 3 rejections the swarm enters `PENDING_APPROVAL` — a person approves or rejects via `POST /swarm/:runId/approve`.



Built on the Azure AI stack, end to end.



Azure AI Foundry REASONING

POWERS ALL FOUR AGENTS

Llama-3.3-70B-Instruct, served serverless on an OpenAI-compatible endpoint. Every agent reasons on the same Azure-hosted model.



Azure Cache for Redis STATE

THE SHARED BLACKBOARD

Holds `SwarmState` per run with a 1800s TTL, so every agent reads and writes one source of truth.



Azure AI Content Safety SAFETY

INTEGRATED AT THE CRITIC LAYER

A configurable safety check the Critic runs before any suggestion is approved.



Custom TS Orchestration

INSPIRED BY SEMANTIC KERNEL HANDOFFS

A hand-built TypeScript agent swarm (`lib/agentSwarm.ts`) modeled on Semantic Kernel handoff patterns — not a framework dependency.

 The orchestration is **original TypeScript** inspired by Semantic Kernel handoff patterns. Reasoning runs **Azure-only** on Azure AI Foundry — no third-party model fallback.



Scale

- **Stateless API** — Fastify scales horizontally; any node serves any request.
- **Resumable runs** — state lives in Redis, so any worker can resume a run.
- **Decoupled workers** — RabbitMQ topic exchange separates the agents from the API.
- **Bounded memory** — per-run TTL caps blackboard growth automatically.
- **Fully containerized** — multi-stage Docker build; `docker-compose` prod stack.

Privacy & Security

- **JWT on every endpoint** — no unauthenticated access to the swarm.
- **No PII in the swarm** — agents pass connection IDs, not personal data.
- **Consent-gated** — only mutually-accepted connections are ever resolvable.
- **Post-quantum E2EE** — messaging uses `PQXDH` + `ML-KEM-768`.
- **Safety + audit** — Critic and Content Safety gate output; every step is traced.

A real run, approved on the first attempt.

AGENT TRACE — GET /SWARM/TRACE/:RUNID

Planner

COMPLETED

6071 ms · intent parsed

Researcher

COMPLETED

2081 ms · 5 connections, 4 venues

Executor

COMPLETED

11108 ms · 3 suggestions

Critic

APPROVED

3995 ms · APPROVED (attempt 1)

GROUNDING SUGGESTIONS RETURNED

Pooja

100 Feet Road, Chattarpur

Sat · 3pm

Geeta

Radisson, MG Road

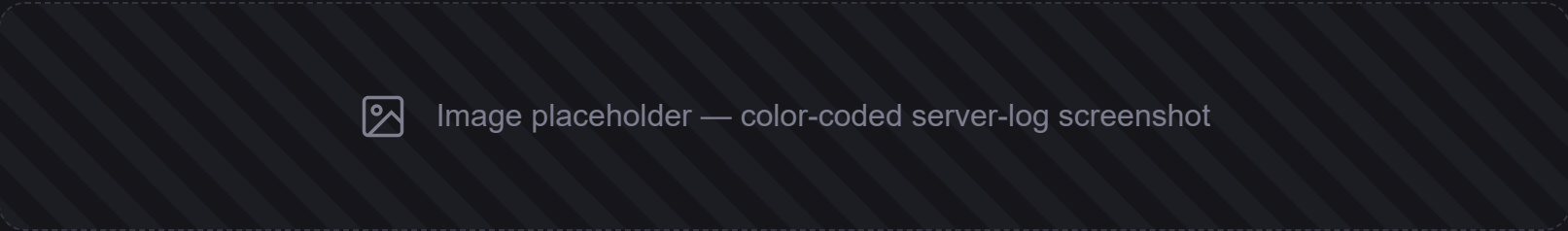
Next Sun

Yash

Wine Shop, MG Road Border

Fri · 6pm

Live run via Azure AI Foundry (Llama-3.3-70B), in Docker. Venues are midpoint-grounded and **approved on attempt 1** — no human override needed.



Where the swarm goes next.

01

AKS deployment with autoscaling

Move the compose stack to Azure Kubernetes Service; scale workers on demand.

03

Per-tenant Azure AI Foundry

Isolate model deployments per tenant for dedicated capacity and governance.

05

Azure Monitor / App Insights

Full observability of agent runs — latency, attempts, and rejection rates.

02

Parallel agents + streaming traces

Run agents concurrently and stream AgentTrace over SSE to a live dashboard.

04

A Memory agent

Add long-term context so suggestions improve across runs and conversations.

**A solo-built, production-grade, fully Dockerized
agent swarm that makes AI social suggestions safe,
specific, and accountable — on the Microsoft stack.**

**"Lokaal: the swarm that earns trust before it makes a
suggestion."**

Akash Pradhan · Solo builder

github.com/akashpradhan439

Microsoft Build AI Hackathon — Agent Swarm